



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

*Faculty of Computing, Universiti Teknologi Malaysia, Johor Bahru,
Malaysia.*

System Analysis and Design (SECD2613)

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Phase 1- Project Proposal and Planning (11%)

Nurul Nasrahtul Balqis Binti Mohamad Fazli	A24CS0177
Zahin Irdina Binti Mohd Zabidy	A24C20216
Salwa Najiha binti Ali Badron	A24CS0183
Siti Nur Farihin Binti Habib Ismail	A24CS0191

Table of Contents

1.0 Introduction.....	3
2.0 Background Study.....	4
3.0 Problem Statement.....	5
4.0 Proposed Solution.....	6
Technical Feasibility.....	7
Operational Feasibility.....	7
Economic Feasibility (Cost-Benefit Analysis).....	7
5.0 Objectives.....	8
6.0 Scope of Project.....	9
7.0 Project Planning.....	10
1. Project Initiation.....	11
1.1. Identify client and stakeholders	
1.2. Conduct stakeholder interviews	
1.3. Define project scope and objectives	
1.4. Project approval.....	11
2. Requirements Analysis.....	11
3. System Design.....	11
4. Development.....	11
5. Testing.....	12
5.1. Unit testing	
5.2. Integration testing	
5.3. System testing.....	12
6. Deployment & Maintenance.....	12
Identify client and stakeholders.....	13
Conduct stakeholder interviews.....	13
Define project scope and objectives.....	13
Project approval.....	13
Unit testing.....	13
Integration testing.....	13
System testing.....	13
8.0 Benefit and Overall Summary of Proposed System.....	16

1.0 Introduction

In this era of rapid technological advancement, digital solutions are continuously transforming the way we live, work, and study. Educational institutions are no exception, and Universiti Teknologi Malaysia (UTM) is actively progressing toward its goal of becoming a fully connected, smart campus. As part of this transformation, enhancing campus services through digital solutions has become a key priority particularly in areas that directly affect student experience and daily convenience.

Despite the progress, many students still face challenges in managing their daily commute within the campus. One of the most persistent challenges faced by students is the lack of efficient, real-time information regarding campus shuttle services. Commuting between faculties, residential colleges, and facilities often becomes unpredictable due to the absence of live tracking or accurate schedules. Many students report frustration from long waits at bus stops, missed shuttles, and difficulty planning their routes, especially during peak hours. This inefficiency not only wastes time but also impacts academic punctuality and overall campus mobility.

To address these issues, our team, Tech Pioneers initiated the SmartShuttle project under UTMFleet. This project aims to develop a smart, user-friendly platform that provides real-time shuttle tracking, accurate bus schedules, and up-to-date route information. This system empowers users to manage their transportation effectively and make informed decisions. The system also benefits shuttle operators, providing them with tools to monitor operations and optimize route planning.

This proposal presents our project development plan, focusing on improving transportation reliability, enhancing user experience, and supporting UTM's smart campus ecosystem. Our strategy involves understanding the current pain points of users, designing a simple yet powerful interface, and building a solution that not only solves existing problems but also lays the groundwork for future enhancements. By applying systematic analysis and design methodologies, we aim to build a platform that not only improves the commuting experience but also contributes to UTM's vision of a connected, intelligent campus environment.

2.0 Background Study

Damia Zulaikha, a first-year engineering student at Universiti Teknologi Malaysia, begins her day early, preparing to attend a morning lecture at the Faculty of Chemical & Energy Engineering. She stays at Kolej Tun Hussein Onn (KTHO), a residential college situated quite far from her faculty which is located up the hill, making it even more difficult to reach on foot, especially during hot or rainy weather. Like many other students, Damia relies heavily on the UTM shuttle bus to move around campus, a service designed to simplify mobility between faculties, colleges, and other key areas.

However, Damia often finds herself waiting at the bus stop without any idea of when the next shuttle will arrive. The only available guide is a printed schedule showing the shuttle route and departure time from the first checkpoint, but it lacks information on estimated arrival times for individual stops such as hers.

Sometimes she ends up waiting 15 to 20 minutes with no sign of a bus. Other times, the shuttle passes by just as she's walking away. This lack of predictability causes frustration and often results in her being late for classes or meetings. As a new student still adjusting to university life, the unreliability of the shuttle system adds unnecessary stress to her daily routine.

In response to this recurring challenge, our team has partnered with UTMFleet to develop the Smart Shuttle System, a mobile application aimed at improving campus transportation. The app will enable students like Damia to track shuttle buses in real time, view estimated arrival times for each stop, and receive instant updates about delays or route changes.

By making the shuttle system more transparent and user-friendly, we aim to enhance the commuting experience for students, staff, and visitors alike. The Smart Shuttle System not only reduces waiting time and stress but also supports UTM's vision of becoming a smart, technology-driven campus where innovation simplifies everyday life.

3.0 Problem Statement

1. Problem 1: Lack of real-time bus tracking and information

Currently, UTM students face significant challenges in planning their daily commute due to the absence of a centralized platform that provides real-time shuttle tracking. Without accurate and up-to-date information on shuttle locations, estimated arrival times, and routes, students often experience long waiting periods, missed trips, and scheduling conflicts. This not only leads to frustration but also affects their punctuality and productivity.

2. Problem 2: Inefficient trip planning and time management

The unpredictability of the existing shuttle system makes it difficult for students to plan their trips efficiently. Many students are left to rely on assumptions or crowd-sourced updates, which are often outdated or inaccurate. This uncertainty in transportation timing hampers effective time management and adds unnecessary stress to students' daily routines.

3. Problem 3: Absence of a unified and user-friendly platform

While shuttle schedules may occasionally be shared through various channels or bulletin boards, there is no centralized, dedicated platform that integrates real-time updates, route information, and scheduling. This lack of integration causes confusion, reduces user engagement, and ultimately discourages students from relying on the campus shuttle system as a dependable mode of transport.

4.0 Proposed Solution

To address the challenges faced by UTM students we propose the development of Smart Shuttle mobile application to help students with campus transportation management issues like lengthy wait times, missed shuttle trips and trouble scheduling commutes. The development of a smart shuttle app or website that provides real-time bus tracking, accurate schedules, and route information. With the use of this app, students could effectively plan their bus trip.

In order to improve user experience and meet campus community mobility needs, the suggested Smart Shuttle system includes a variety of important features. Firstly, students will be able to monitor the exact position and estimated arrival time of shuttle buses on various routes through the app's real-time tracking feature. This will help students avoid wasting time waiting and make it easier for them to decide when and where to catch the shuttle.

Additionally, the application will feature dynamic route planning and up-to-date schedules. Users can efficiently plan trips between academic buildings, residential colleges, and campus facilities by viewing full details on all available shuttle routes and stops. To ensure students are informed at all times, the app will also send out notifications schedule modifications, delays, or service interruptions.

With the goal to ensure that students can easily get the information they require without having to navigate through intricate menus, the interface will be developed with accessibility and simplicity in mind. The software will allow effective use of the campus shuttle system and appeal to both new and returning students with its clear, user-friendly style.

A feedback and reporting tool will also be included in the app, allowing students to comment on shuttle services, identify problems such as delays or crowding, and recommend improvements. By promoting interaction between users and campus transport services, this feature makes it possible to continuously enhance the service in response to student needs.

Technical Feasibility

The Smart Shuttle app can be developed using tools and technologies that are already available, like mobile app frameworks, GPS tracking, and map services such as Google Maps. We will need skilled developers to build and maintain the app. For the app to function properly, it will require an effective framework to handle shuttle data, routes, scheduling, and user accounts. Additionally, it's important to ensure that the app is user-safe, flexible to accommodate future users, and simple to update and maintain.

Operational Feasibility

For the Smart Shuttle app to run smoothly, we will need the right people to help manage and support it after it is launched. This includes making sure that the GPS tracking functions well, that the data is maintained up to date, and that students receive help if they encounter any issues with the app. To make it easier for students to understand how to use the app, we also want to offer user manuals and videos. We will be able to fix any issues and enhance the app with the help of student input.

Economic Feasibility (Cost-Benefit Analysis)

The cost of developing the Smart Shuttle app will cover things like app design, GPS tracking, hosting, maintenance, and user support to keep it running smoothly. To help cover these costs, we can include ads from campus services or local businesses and form partnerships that benefit both sides. We will also carry out a cost-benefit analysis to make sure the app is a good investment and brings real value to students and the university by improving campus transportation.

5.0 Objectives

The Smart Shuttle System is designed with several key objectives to enhance transportation efficiency and convenience for students and staff at UTM's Skudai campus.

1. **Develop and implement a real-time tracking system** using GPS technology and interactive map-based navigation tools. This feature will allow users to monitor shuttle locations accurately, ensuring they can make informed decisions about their travel routes.
2. **Improve time management for passengers by providing up-to-date shuttle schedules and real-time estimated arrival times.** By offering precise and reliable information, the system helps users minimize unnecessary waiting periods, allowing them to better plan their daily commutes. This is particularly beneficial in reducing the risk of missed classes, appointments, or meetings due to transportation delays.
3. **Optimize shuttle operations by analyzing demand patterns and service conditions.** By ensuring that shuttle availability aligns with actual usage, the university can enhance resource efficiency while providing a more responsive transportation service.
4. **Integrating advanced digital solutions into essential university services.** Through this project, UTM reinforces its commitment to innovation, sustainability, and improved quality of life for its campus community. By leveraging modern technology, the system not only streamlines transportation but also contributes to a more connected and digitally empowered campus environment.

6.0 Scope of Project

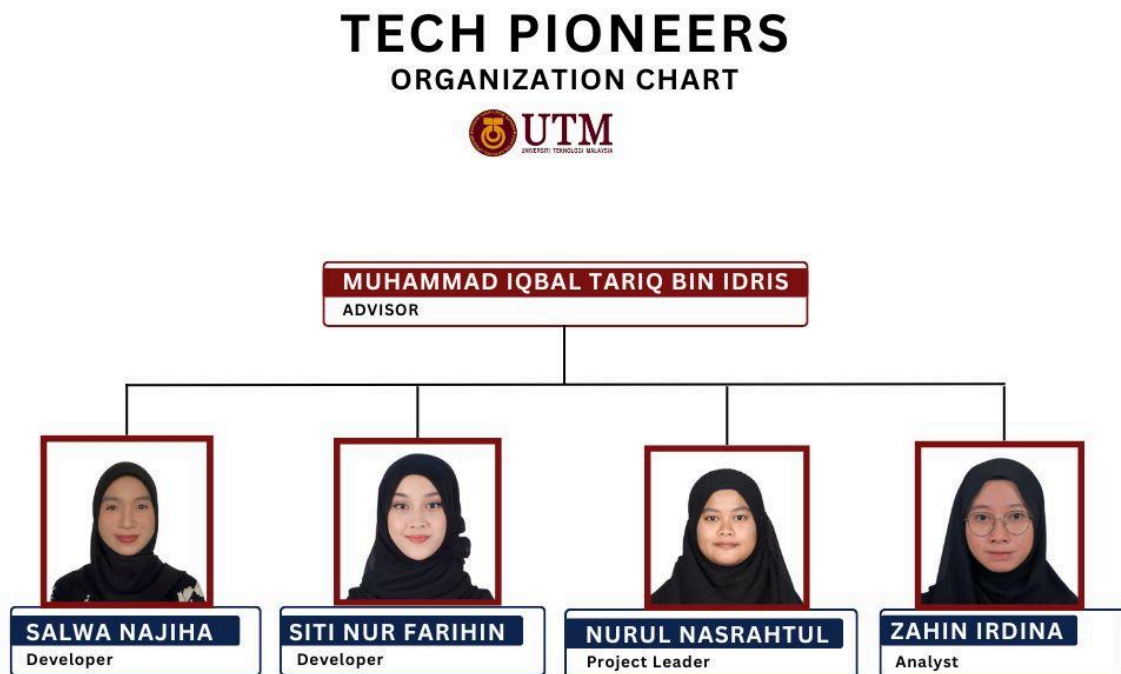
The Smart Shuttle System is designed to enhance the transportation ecosystem within the UTM Skudai campus. Its scope is carefully defined to ensure the project remains manageable while delivering significant impact. The system will include the development of a mobile application compatible with both Android and iOS platforms, providing users with easy access to shuttle services.

Key features of the system include GPS-based shuttle tracking, which will be displayed on an interactive campus map, allowing users to monitor shuttle locations in real time. Additionally, the app will provide real-time schedule data, including estimated arrival and departure times, to help users plan their journeys efficiently. To ensure secure access, the system will incorporate user authentication and session management, including a login system for both administrators and regular users.

For administrative oversight, a web-based dashboard will be implemented, enabling system administrators to manage shuttle operations effectively. To keep users informed, the system will also support push notifications and alerts regarding shuttle-related updates, such as delays or schedule changes. Together, these features aim to create a seamless and reliable transportation experience for the UTM Skudai campus community.

7.0 Project Planning

7.1 Human Resource



7.2 Work Breakdown Structure

1. Project Initiation

- 1.1. Identify client and stakeholders
- 1.2. Conduct stakeholder interviews
- 1.3. Define project scope and objectives
- 1.4. Project approval

2. Requirements Analysis

- 2.1. Gather functional and non-functional requirements
- 2.2. Analyze existing shuttle system
- 2.3. Create user stories and use cases
- 2.4. Develop project plan
- 2.5. Client approval on requirements

3. System Design

- 3.1. Design system architecture (frontend, backend, GPS integration)
- 3.2. Design UI/UX wireframes and prototypes
- 3.3. Design database schema
- 3.4. API & integration planning
- 3.5. Security Framework

4. Development

- 4.1. Backend development
 - 4.1.1. Setup server and database
 - 4.1.2. API development
 - 4.1.3. Authentication & authorization
 - 4.1.4. Shuttle tracking logic
- 4.2. Mobile app development
 - 4.2.1. Login/Register UI
 - 4.2.2. Map and real-time tracking
 - 4.2.3. Route and schedule display
 - 4.2.4. Notification system

4.3. Admin dashboard (Web)

4.3.1. Shuttle and route management

4.3.2. Schedule editor

4.3.3. User management

5. Testing

5.1. Unit testing

5.2. Integration testing

5.3. System testing

5.4. User acceptance testing (UAT) with stakeholders

5.5. Bug fixing and iteration

6. Deployment & Maintenance

6.1. Deploy to production server

6.2. Deploy backend and database

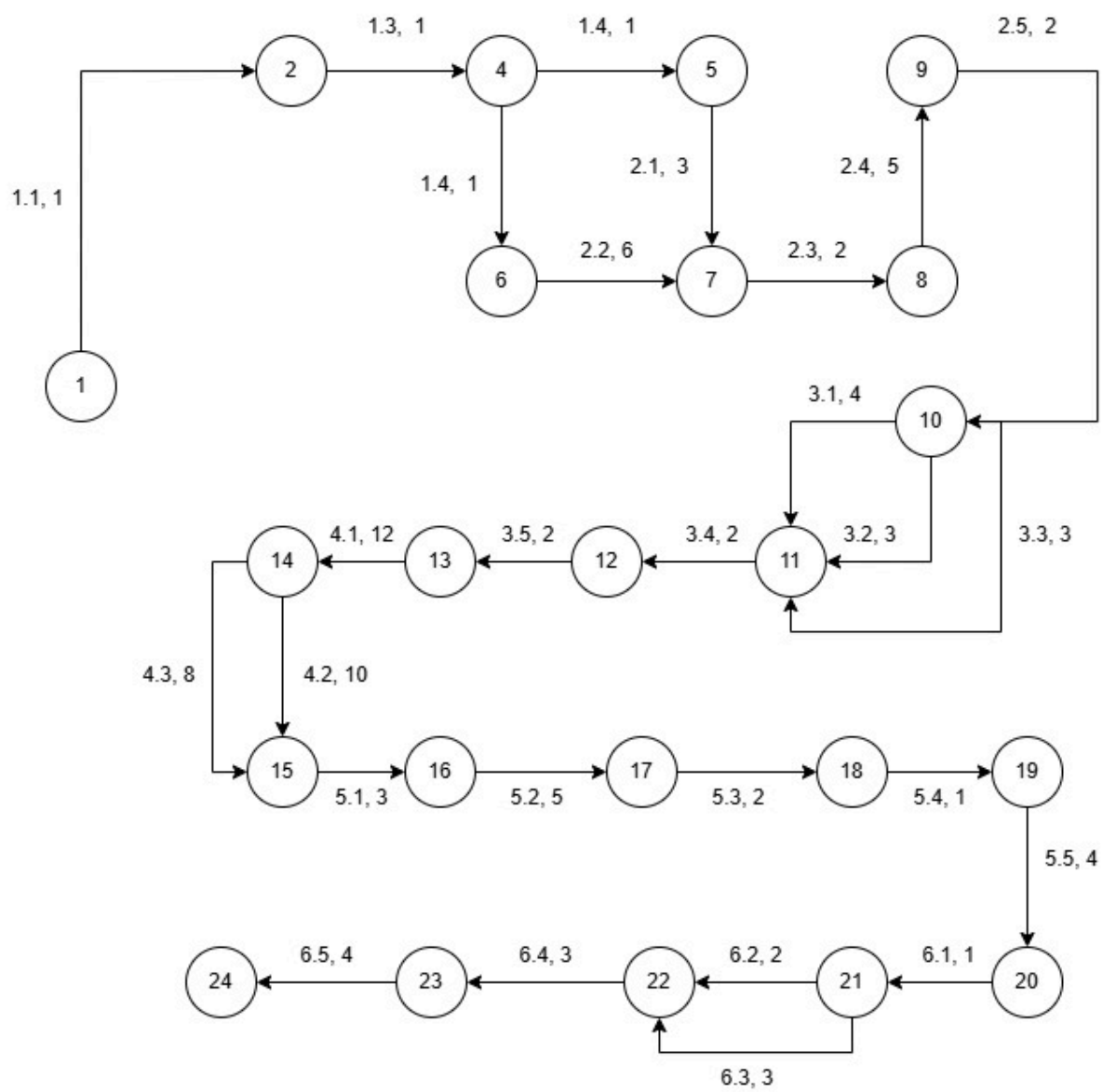
6.3. Publish mobile app

6.4. User onboarding and training

6.5. Ongoing Bug Fixes & Support

7.3 PERT Chart

	Activity	Predecessor	Duration
1.1	Identify client and stakeholders	None	1
1.2	Conduct stakeholder interviews	1.1	1
1.3	Define project scope and objectives	1.1	1
1.4	Project approval	1.3	1
2.1	Gather functional and non-functional requirements	1.4	3
2.2	Analyze existing shuttle system	1.4	6
2.3	Create user stories and use cases	2.1, 2.2	2
2.4	Develop project plan	2.3	5
2.5	Client approval on requirements	2.4	2
3.1	Design system architecture (frontend, backend, GPS integration)	2.5	4
3.2	Design UI/UX wireframes and prototypes	2.5	3
3.3	Design database schema	2.5	3
3.4	API & integration planning	3.1, 3.2, 3.3	2
3.5	Security Framework	3.4	2
4.1	Backend development	3.4, 3.5	12
4.2	Mobile app development	4.1	10
4.3	Admin dashboard (Web)	4.1	8
5.1	Unit testing	4.2, 4.3	3
5.2	Integration testing	5.1	5
5.3	System testing	5.2	2
5.4	User acceptance testing (UAT) with stakeholders	5.3	1
5.5	Bug fixing and iteration	5.4	4
6.1	Deploy to production server	5.5	1
6.2	Deploy backend and database	6.1	2
6.3	Publish mobile app	6.1	3
6.4	User onboarding and training	6.2, 6.3	3
6.5	Ongoing Bug Fixes & Support	6.4	4

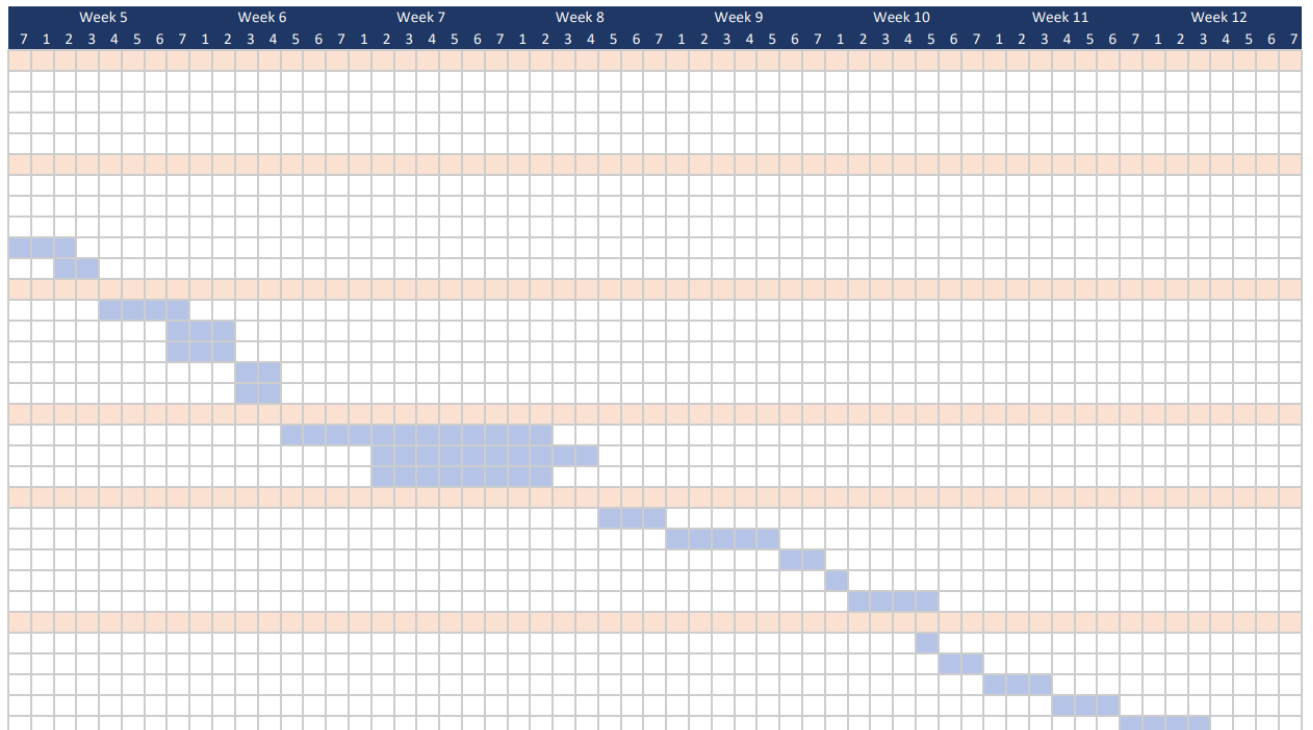
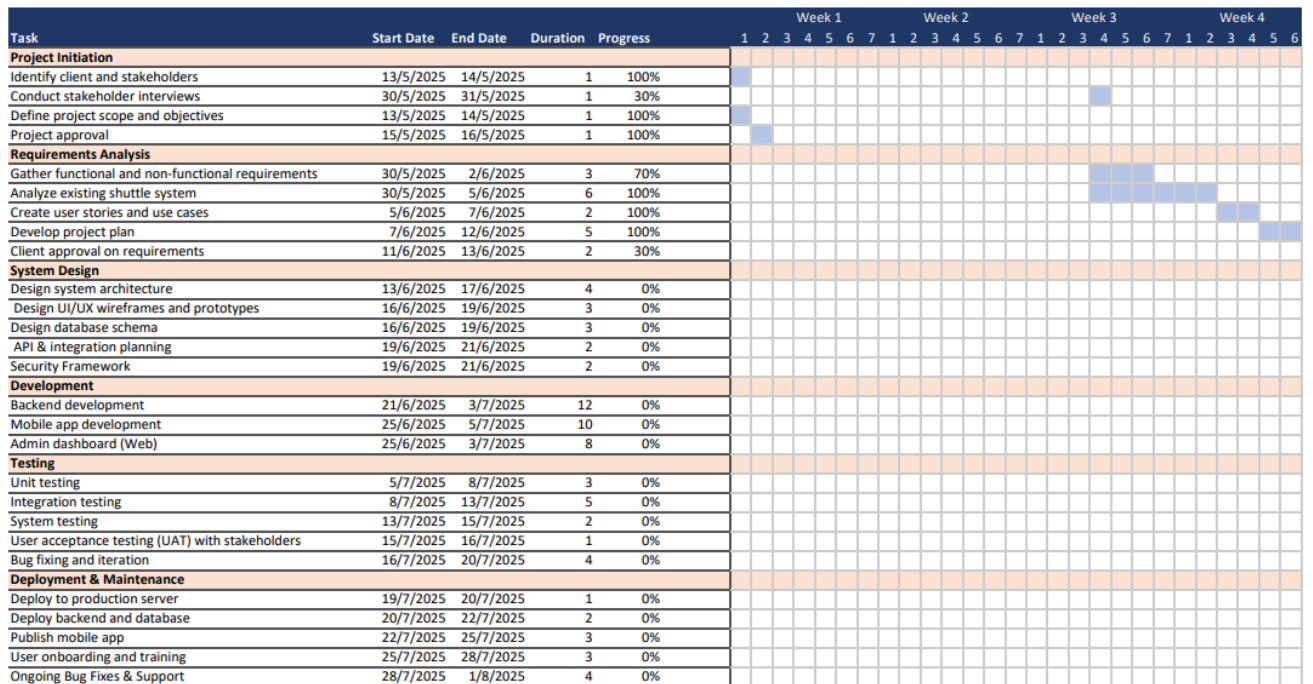


7.4 Gantt Chart

UTM SMART SHUTTLE

COMPANY NAME: TECH PIONEERS

DATE: 13/5/2025



8.0 Benefit and Overall Summary of Proposed System

The implementation of the Smart Shuttle System at UTM's Skudai campus is anticipated to bring significant improvements to campus transportation, benefiting both users and administrators. One of the most notable advantages is the expected increase in satisfaction among students and staff. By providing reliable, real-time shuttle information, the system will reduce uncertainty and anxiety related to transportation delays, fostering greater trust in campus shuttle services. This enhanced reliability will contribute to a more positive overall campus experience.

Additionally, the system will save valuable time for daily commuters. With live GPS tracking and accurate schedule updates, users will be able to plan their movements more efficiently, minimizing unnecessary waiting periods and ensuring punctuality for classes, meetings, and other campus activities. This time optimization will help students and staff maintain better daily routines, ultimately supporting academic and professional productivity.

From an operational perspective, the Smart Shuttle System will provide administrators with improved control over shuttle services. Through the web-based management dashboard, they will be able to monitor shuttle performance in real time, identify peak demand periods, and make data-driven decisions to optimize scheduling and resource allocation. This will lead to more efficient shuttle operations, reducing overcrowding during busy hours and ensuring better service availability when needed.

Furthermore, the system will enhance overall campus mobility by removing transportation barriers. Students and staff will no longer face difficulties in reaching classes, libraries, administrative offices, or other key facilities, as the shuttle service will be more predictable and accessible. This improvement in connectivity will support academic success and streamline daily campus life.

Lastly, the Smart Shuttle System aligns with UTM's broader Smart Campus vision, demonstrating the university's commitment to digital transformation and innovation. By integrating modern technologies into essential services, UTM reinforces its position as a forward-thinking institution that prioritizes sustainability, efficiency, and user convenience. The project not only modernizes campus transportation but also sets a foundation for future smart initiatives, contributing to a more connected and technologically advanced university environment.

Overall, the Smart Shuttle System promises to deliver a more efficient, transparent, and user-friendly transportation solution, benefiting the entire UTM community while supporting the university's long-term strategic goals.